

# All-vdW ferroelectric and multiferroic tunnel junctions for novel memory hardware

Cheng Gong

Department of Electrical and Computer Engineering, University of Maryland, College Park, MD 20742, USA, gongc@umd.edu

Ferroelectric tunnel junctions (FTJs) and magnetic tunnel junctions (MTJs) have been the mainstream non-volatile memory devices underpinning modern information technologies. Oxides are commonly employed as spacer layers in these devices and epitaxial growth has been the common approach to synthesize the multilayer junctions. However, remaining issues are still outstanding – for example, defects are unavoidable, Fermi level pinning is always present, and lattice match between dissimilar materials is demanded. This Digest introduces our recent development of all – van der Waals (vdW) ferroelectric and multiferroic tunnel junctions (MFTJs). By eliminating defects and Fermi level pinning, with unlimited freedom in assembling dissimilar vdW layers, all-vdW FTJs and MFTJs are shown to exhibit superior memory device performance and unparalleled flexibility in junction construction.

**Index Terms**—Ferroelectric, multiferroics, tunnel junctions, van der Waals materials.

## I. INTRODUCTION

FERROELECTRIC and magnetic tunnel junctions are two critical types of non-volatile memory devices. Ferroelectric tunnel junctions (FTJs) comprise of two unidentical metal electrodes and a nanometer-thin ferroelectric spacer. When the ferroelectric polarization is switched by an external electric field, the tunneling barrier profile between the two electrodes is altered, leading to a changed electrical resistance. This phenomenon is coined as tunneling electroresistance effect (TER) [1]. Magnetic tunnel junctions (MTJs) comprise of two magnetic metal electrodes and a nanometer-thin insulating spacer. When the magnetizations of the two magnetic electrodes are aligned parallel versus anti-parallel, the electrical resistances differ. This phenomenon is coined as tunnelling magnetoresistance effect (TMR). Early theories [2] predicted and experiments also realized trilayer structures of two unidentical magnetic electrodes separated by a nanometer-thin ferroelectric film, which can combine the functions of FTJs and MTJs and give rise to four distinct electrical resistances by switching the ferroelectric polarization and the relative magnetization orientations of the two magnetic layers, respectively. The four-state non-volatile memory devices, coined as multiferroic tunnel junctions (MFTJs), are a new type of multi-state hardware, potentially offering enhanced functions to outperform the traditional binary state memory devices.

Given the trilayer structures of FTJs, MTJs and MFTJs are conventionally synthesized by epitaxy, lattice match between the three layers is a prerequisite. Also, because of the unavoidable defects in thin film oxides, device performance could be compromised by leakage current, Fermi level pinning, etc. The emerging vdW ferroics (both ferroelectrics [3,4] and ferromagnets [5]) may provide new material platforms that (1) can maintain their high crystallinity with negligible defects even down to the atomic-thin regime and (2) can be stacked with dissimilar vdW materials to form multilayer heterostructures without the lattice-match restriction.

## II. VdW FTJs

The ON/OFF ratio and current densities are critical figures of merits of FTJ memory devices. We studied vdW FTJs (see Fig. 1) based on two types of ferroelectrics: larger-bandgap  $\text{CuInP}_2\text{S}_6$  (2.9 eV bandgap) and smaller-bandgap  $\text{In}_2\text{Se}_3$  (1.4 eV bandgap). Generally, we find that the larger-bandgap  $\text{CuInP}_2\text{S}_6$  tends to cause higher ON-OFF ratios of  $10^7$  or above, yet with modest ON-state current densities of approximately  $10^2$ – $10^3$  A/cm<sup>2</sup>. In contrast, the smaller-bandgap  $\text{In}_2\text{Se}_3$  tends to yield higher ON-state current densities of approximately  $10^4$ – $10^5$  A/cm<sup>2</sup>, yet with lower ON/OFF ratios. The above summary is based on FTJs using thin vdW ferroelectrics (i.e., 3–5 nm).

Through tuning the thickness of the smaller-bandgap ferroelectrics  $\text{In}_2\text{Se}_3$ , we have overcome the low ON/OFF ratios [6]. When  $\text{In}_2\text{Se}_3$  thickness is increased 10X, we find the OFF state is substantially suppressed by about 4 orders of magnitude, leading to the enhanced ON/OFF ratio up to the level of  $10^7$  for  $\text{In}_2\text{Se}_3$ -FTJs, which meanwhile maintain high ON-state current densities on the order of  $10^4$  A/cm<sup>2</sup>.

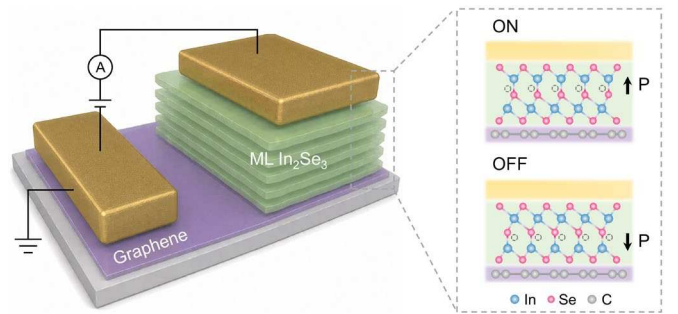


Fig. 1. Illustration of  $\text{In}_2\text{Se}_3$ -based vdW FTJs.

## III. ALL-VdW MFTJs

We experimentally demonstrate all-vdW MFTJs, with four distinct electrical resistance clearly realized (see Fig. 2) [7]. We engineer the electrode asymmetry to enhance the ON/OFF ratio through selecting different magnetic electrode layers.

Furthermore, we engineer the spacer layers to adjust the ON/state current density by selecting larger-bandgap  $\text{CuInP}_2\text{S}_6$  or smaller-bandgap  $\text{In}_2\text{Se}_3$ . The wide tunability enabled by freely choosing the materials and thicknesses of all the three layers demonstrates the fully tailorable all-vdW MFTJs. Such a wide tunability has never been possible in the synthetic oxide multilayer systems.

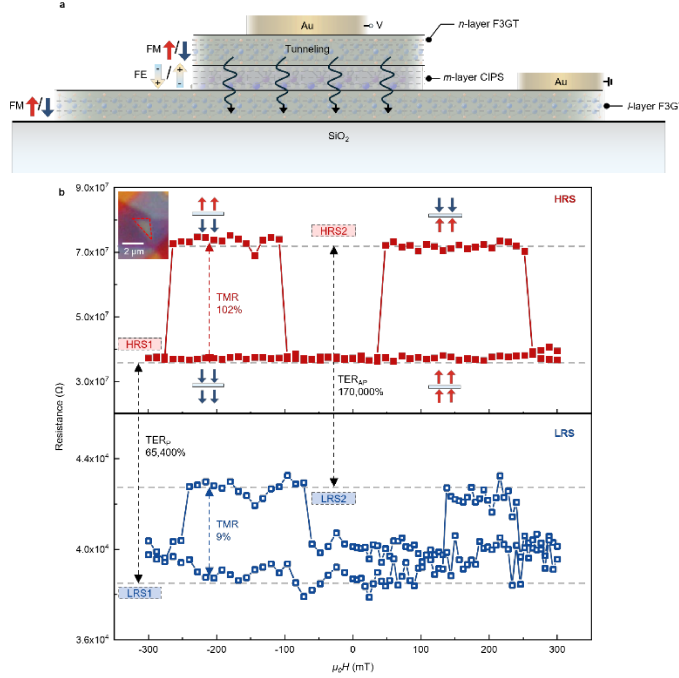


Fig. 2. Illustration and demonstration of all-vdW MFTJs consisting of a pair of unidentical  $\text{Fe}_3\text{GeTe}_2$  electrodes and a  $\text{CuInP}_2\text{S}_6$  ferroelectric spacer.

#### IV. CONCLUSIONS

This Digest introduces our recent development of vdW FTJs and all-vdW MFTJs. By tuning the ferroelectric materials and thicknesses, we have reached high ON/OFF ratios and high ON-state current densities simultaneously in  $\text{In}_2\text{Se}_3$ -based vdW FTJs. By selecting the ferroelectric spacer and the ferromagnetic electrodes, we have demonstrated the all-vdW MFTJs with four clearly resolved distinct electrical resistances with highly tunable device performance.

#### REFERENCES

- [1] M. Ye, Zhuravlev, R. F. Sabirianov, S. S. Jaswal, and E. Y. Tsymlal, "Giant electroresistance in ferroelectric tunnel junctions," *Phys. Rev. Lett.*, vol. 94, art. 246802, Jun. 2005.
- [2] J. P. Velev, C.-G. Duan, J. D. Burton, A. Smogunov, M. K. Niranjan, E. Tosatti, S. S. Jaswal, and E. Y. Tsymlal, "Magnetic tunnel junctions with ferroelectric barriers: prediction of four resistance States from first principles," *Nano Lett.*, vol. 9, pp. 271–350, Dec. 2008.
- [3] J. Wu, H.-Y. Chen, N. Yang, J. Cao, X. Yan, F. Liu, Q. Sun, X. Ling, J. Guo, and H. Wang, "High tunnelling electroresistance in a ferroelectric van der Waals heterojunction via giant barrier height modulation," *Nat. Electron.*, vol. 3, pp. 466–472, Jul. 2020.
- [4] W. Ding, J. Zhu, Z. Wang, Y. Gao, D. Xiao, Y. Gu, Z. Zhang, and W. Zhu, "Prediction of intrinsic two-dimensional ferroelectrics in  $\text{In}_2\text{Se}_3$  and other  $\text{III}_2\text{-VI}_3$  van der Waals materials," *Nat. Commun.*, vol. 8, art. 14956, Apr. 2017.
- [5] C. Gong, L. Li, Z. Li, H. Ji, A. Stern, Y. Xia, T. Cao, W. Bao, C. Wang, Y. Wang, Z. Q. Qiu, R. J. Cava, S. G. Louie, J. Xia, and X. Zhang, "Discovery of intrinsic ferromagnetism in two-dimensional van der Waals crystals," *Nature*, vol. 546, pp. 265–269, Apr. 2017.
- [6] Y. Sun, et al. "Reconciliation of diametric transport regimes enables high-performance van der Waals ferroelectric heterojunctions," *Nat. Electron.*, submitted for publication.
- [7] T. Xie, Q. Wang, H. Zhang, K. Acharya, J. Chen, C. Liu, Z. Song, S. A. Deitemyer, H. S. Arachchige, Q. Tan, A. F. May, S. H. Lee, M. A. Susner, Z. Mao, M. A. McGuire, X. Ling, D. Mandrus, X. Zhang, S.-J. Gong, T. R. Paudel, R. Ramesh, E. Y. Tsymlal, and C. Gong, "Tailorable multiferroic tunnel junctions from all-van der Waals multilayer stacking," *Nat. Nanotechnol.*, vol. 21, pp. 366–373, Feb. 2026.